

**From Association to Causation:
Multi-Method Propensity Score Analysis of
Physical Activity and Cardiovascular
Disease with Identification Diagnostics and
Mediation**

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Abstract

Under the potential-outcomes framework, I use the Ulianova public cardiovascular dataset to estimate the causal effect of physical activity on cross-sectional cardiovascular disease (CVD) prevalence. Guided by a directed acyclic graph (DAG) and the backdoor criterion, I specify an adjustment set and triangulate inverse probability weighting (IPW), g-formula, doubly robust augmented IPW (AIPW), propensity score (PS) matching and stratification, and linear probability models (LPM). I systematically assess propensity score overlap (positivity), covariate balance, BMI mediation decomposition, adjustment-set sensitivity, and E-value bounds for unmeasured confounding. After controlling for major confounders (age, sex, smoking, alcohol, cholesterol, glucose, and BMI), physical activity shows a consistent protective association: IPW-adjusted prevalence difference = -0.044 (95% CI $[-0.053, -0.035]$), NNT ≈ 23 on the prevalence scale; all seven identification–estimation strategies converge in direction. Protection operates mainly through a direct path not mediated by BMI (indirect share $\approx 5.6\%$). Because the data are cross-sectional and observational, reverse causation and residual confounding cannot be ruled out; conclusions should be read as cross-sectional supportive rather than confirmatory evidence.

Keywords: physical activity; cardiovascular disease; causal inference; propensity score; inverse probability weighting

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1 Background and Research Questions

1.1 Field Context and Prior Evidence

Globally, insufficient physical activity and sedentary behavior continue to rise and are widely recognized as modifiable risk factors for noncommunicable diseases, especially cardiovascular disease (CVD) [1]. Mi et al. [1], in a *Circulation Research* review, note that despite decades of accumulating epidemiological evidence and public health guidelines promoting targets such as at least 150 minutes per week of moderate-intensity aerobic activity, large segments of the U.S. and global populations still fail to meet recommended activity levels, with disproportionately low activity among racial/ethnic minorities, women, and older adults. Mechanistically, regular physical activity may protect the cardiovascular system by improving endothelial function, reducing systemic inflammation, optimizing blood pressure and lipid profiles, and increasing cardiorespiratory fitness [1]. Thus, whether increasing activity lowers CVD prevalence is a core public health question and a canonical setting for causal inference methods.

Observational studies provide substantial supportive evidence. Garcia et al. [2] conducted a dose–response meta-analysis of 94 prospective cohorts with over 30 million participants and found nonlinear inverse associations between non-occupational physical activity and all-cause mortality, CVD mortality, and several cancer outcomes: the largest risk reductions occur from “inactive” to roughly 8.75 mMET-hours per week (equivalent to guideline-recommended 150 minutes/week of moderate-to-vigorous aerobic activity). At that dose, the relative risk (RR) of CVD death is approximately 0.71 (95% CI: 0.66–0.77) and all-cause mortality RR is approximately 0.69; if all insufficiently active individuals reached that level, about 15.7% of premature deaths could be avoided. Mi et al. [1] further summarize that higher activity is consistently associated with lower mortality and risk of coronary heart disease, ischemic stroke, heart failure, atrial fibrillation, and other CVD subtypes across age, sex, and baseline health status; pooled analyses of over 2 million participants with questionnaire-assessed activity link meeting guideline activity with roughly 22% lower mortality, while accelerometer-based studies often report larger effect sizes. Mechanistic literature suggests that activity’s impact on CVD is not captured by a single “total effect”: Chen et al. found mediation and moderation through body-fat percentage, blood pressure, and lipids among Chinese adults [3]; Wang et al. report that moderate-to-vigorous recreational activity partly affects hypertension through obesity indices [4]; Hall et al. note that cardiorespiratory fitness (CRF) as a mediator can shift activity-intensity thresholds relevant to cardiometabolic health [5]. Together, these results portray a landscape of protective associations and multi-pathway mechanisms, providing literature support for the mediation decomposition in this paper.

1.2 Limitations of Observational Research and Position of This Study

Moving from association to causation, observational PA–CVD studies face structural limitations that motivate this project. Activity is not randomly assigned: younger, metabolically healthier, and socioeconomically advantaged individuals tend to be more active, and these factors also influence CVD diagnosis, so classical confounding prevents crude comparisons from receiving a causal interpretation [1,6]. In cross-sectional or short-follow-up designs, reverse causation is equally problematic—individuals with existing CVD symp-

toms or diagnoses may reduce activity, inflating the appearance that active individuals have lower prevalence [1]. When activity is measured by self-report questionnaires or coarse categories, recall bias and misclassification further weaken credibility; Mi et al. [1] note that questionnaire and accelerometer estimates for the same activity dose can differ severalfold, with varying measurement reliability across devices and populations (e.g., older adults). Even Mendelian randomization (MR) designs do not guarantee robust causal conclusions: Qiu et al. [7], in a two-sample MR study based on European GWAS, found only vigorous physical activity (VPA) showed a modest causal association with major coronary heart disease (IVW OR= 0.95, 95% CI: 0.91–0.99), while average activity and other intensity subtypes showed no substantial causal links with eight chronic diseases—suggesting limited genetic instrument representation of “general activity” and that MR cannot simply be equated with “no causal effect of activity on CVD,” but does show causal evidence strength varies by design. Mi et al. [1] also emphasize that direct RCT evidence that PA lowers mortality and CVD remains relatively limited; long-term adherence and ethical constraints make large randomized trials difficult. In sum, the field has established a protective direction and dose clues for PA, but on a single observational dataset there remains room for methodological integration and transparent reporting of how to estimate interpretable average causal effects (ACE) under explicit causal assumptions, diagnose identification conditions, decompose mechanisms, and assess residual confounding.

Existing high-quality evidence predominantly relies on prospective cohort designs, focusing on the long-term causal effects of PA on CVD incidence and mortality; standardized causal estimation, identification diagnostics, and mechanism decomposition targeting population prevalence levels remain relatively uncommon. This study leverages a large cross-sectional examination dataset to estimate, under the potential-outcomes framework, the average causal effect of PA on CVD prevalence, and systematically conducts positivity checks, covariate balance diagnostics, and sensitivity analyses.

Against this background, I conduct a complete observational causal analysis pipeline using the Ulianova public cardiovascular dataset [8]. Compared with existing literature, I do not claim new biological findings or population-level causal certainty; I focus on a methodologically transparent, reproducible estimation and diagnostic chain under explicit assumptions: under DAG and backdoor guidance, the estimand is $ACE = \mathbb{E}[Y(1) - Y(0)]$ (because the outcome is cross-sectional CVD prevalence, this is a prevalence difference rather than a cohort incidence risk difference), distinguishing total-effect adjustment from mediation decomposition with respect to BMI [6, 11]; on one dataset I triangulate IPW, g-formula, doubly robust AIPW, linear models, and PS matching/stratification [6, 9, 13, 14]; I systematically report positivity and covariate balance diagnostics [10]; and under the VanderWeele framework I decompose BMI mediation paths, combining adjustment-set sensitivity, effect modification, and E-value analysis [11, 12]. Section 2 states identification assumptions and estimation principles; Section 3 reports multi-method ACE; Sections 4–5 test positivity and balance—only when estimates converge and diagnostics pass do subsequent mechanism and sensitivity discussions become meaningful.

1.3 Data, Estimand, and Causal Structure

Data come from the Kaggle Ulianova cardiovascular examination dataset [8] (original file `cardio_train.csv` with 70,000 records). Before analysis, I exclude 1,608 observations with implausible blood pressure ($ap_hi \in [80, 200]$, $ap_lo \in [40, 120]$, and $ap_hi > ap_lo$),

BMI (15–60), or height (140–220 cm), retaining $N = 68,392$ valid samples. Note: other small-sample files in the same directory (e.g., UCI Heart Disease) are not used in this study. Outcome Y is CVD diagnosis (prevalence 49.4%); treatment A is binary self-reported physical activity (80.3% active). Covariates include age, sex, smoking, alcohol use, cholesterol, glucose, and BMI, matching the multidimensional L structure required for propensity score/IPW methods, with variables measured consistently in the same examination workflow. Importantly, the data are cross-sectional, activity is a coarse self-reported category, intensity and duration cannot be distinguished, objective measurement is unavailable, and reverse causation cannot be ruled out; conclusions should be understood as causal estimation under stated assumptions on this dataset, not a simple replication of Garcia et al. [2] or other large prospective meta-analyses.

Under the potential-outcomes framework, let $A_i = 1$ denote individual i is physically active, $Y_i = 1$ denote CVD presence, and $Y_i(a)$ the potential outcome under treatment a . The target estimand is the average causal effect

$$\text{ACE} = \mathbb{E}[Y_i(1) - Y_i(0)].$$

If $\text{ACE} < 0$, intervening to move the entire population from inactive ($A = 0$) to active ($A = 1$) would reduce average CVD prevalence by $|\text{ACE}|$. This differs from an incidence risk difference in cohort studies: cross-sectional prevalence mixes prevalent disease with current risk, so causal extrapolation requires extra caution [6]. It also differs from an associational comparison of currently active versus inactive individuals, which mixes causal effects with selective entry into activity.

Figure 1 shows the causal structure. Age, sex, smoking, alcohol use, cholesterol, and glucose affect both activity and CVD prevalence, forming backdoor paths $A \leftarrow L \rightarrow Y$; BMI likewise, and may biologically serve partly as mediator $A \rightarrow M \rightarrow Y$ [3, 4]. Unmeasured factors U (e.g., family history, diet, healthcare access) that affect both A and Y constitute residual confounding that cannot be directly tested. The primary adjustment set is

$$L = \{\text{age, sex, smoking, alcohol, cholesterol, glucose, BMI}\},$$

chosen by the backdoor criterion: after conditioning on L , A is conditionally independent of $(Y(1), Y(0))$ [6, 13]. The main analysis controls BMI as a confounder to estimate total effect; mechanism analysis later removes BMI from L and treats it separately as mediator M to decompose natural direct effect (NDE) and natural indirect effect (NIE) [11]. The main analysis deliberately does not control concurrent blood pressure: in cross-sectional data, blood pressure may reflect long-term activity effects (mediation) or existing cardiovascular pathology (confounding), and including it risks over-adjusting for mediators or conditioning on colliders [6]; robustness analysis separately tests adding blood pressure.

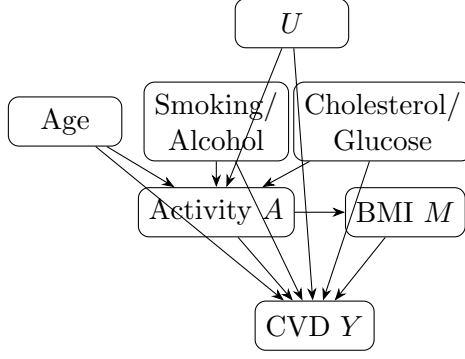


Figure 1: Causal DAG. L includes age, sex, smoking, alcohol, cholesterol, glucose, and BMI (total-effect model); mechanism analysis separates BMI as M from L .

2 Identification Assumptions and Estimation Principles

Identifying ACE from observational data requires assumptions linking potential outcomes to observed variables; only when these hold (or approximately hold) do IPW, g-formula, and propensity score methods receive causal interpretation [6]. This section explains the meaning of identification assumptions and their correspondence in this dataset, then describes how each estimator is constructed and why they should yield similar answers.

2.1 Identification Assumptions

Consistency requires that for each individual, if observed treatment is $A = a$, then observed outcome Y equals potential outcome $Y(a)$, i.e., $Y = Y(A) = \sum_a Y(a)I(A = a)$ [6]. Here, A is binary self-reported activity and Y is CVD diagnosis; consistency implies the definition of “activity” is unambiguous between potential outcomes and observed variables, with no distinct “versions” of treatment for the same A value. Because activity is only a coarse self-reported category, it may encompass different intensities and types; the assumption should be understood as a causal effect of the observed active/inactive label, not of a specific exercise prescription (e.g., 150 minutes MVPA per week) [1]. If activity measurement involves non-differential misclassification (i.e., CVD status does not affect the probability of misreporting activity), classical epidemiology predicts attenuation of effect estimates toward the null—another reason, beyond outcome definition, that our magnitudes may be smaller than those in large prospective cohort studies.

The stable unit treatment value assumption (SUTVA) comprises no interference and treatment-variation irrelevance [6]. No interference requires individual i ’s potential outcomes depend only on A_i , not others’ treatment; in cross-sectional examination data, inter-individual influence on activity is typically weak, so this assumption is relatively plausible. Treatment-variation irrelevance requires $Y(1)$ and $Y(0)$ to be well defined—if “active” means very different activity patterns across individuals, interpretability of potential outcomes declines. Consistency and SUTVA together ensure I estimate a well-defined $\mathbb{E}[Y(1) - Y(0)]$, not an associational contrast confounded by measurement error and definitional drift.

Conditional exchangeability (also ignorability or “no unmeasured confounding”) is the core of observational causal inference: after controlling L , potential outcomes are

independent of treatment,

$$Y(1), Y(0) \perp\!\!\!\perp A \mid L.$$

This means that within subpopulations with the same L , active versus inactive individuals are approximately comparable, as if conditionally randomized within that stratum [6, 13]. In Figure 1, L blocks backdoor paths such as $A \leftarrow L \rightarrow Y$; if additionally $U \rightarrow A$ and $U \rightarrow Y$ with U unmeasured, exchangeability fails in the true data-generating mechanism, and identification proceeds only under the untestable premise that L is sufficient [6]. I limit measured L to age, sex, smoking, alcohol, cholesterol, glucose, and BMI, acknowledging that family history, diet, socioeconomic status, and other U may still threaten identification—Section 7 E-value analysis quantifies how strong such residual confounding would need to be to explain away the estimate [12]. Because the data are cross-sectional, the temporal ordering of exposure and outcome cannot be verified from the data alone; reverse causation may threaten conditional exchangeability (see Section 8.2 for discussion).

Positivity (overlap) requires that on the support of L ,

$$0 < \mathbb{P}(A = 1 \mid L = l) < 1,$$

i.e., for each covariate profile, both active and inactive individuals exist [6, 10]. If for some l , $\mathbb{P}(A = 1 \mid L = l) \approx 1$, then $\mathbb{E}[Y(0)]$ lacks empirical support in that stratum, IPW weights $1/(1 - e(L))$ explode, and matching finds no controls. Positivity differs from exchangeability: the former is directly diagnosable via propensity score distributions and balance plots (Sections 4–5), the latter cannot be proven from data. With 80.3% active in this sample, positivity faces the reality of high treated proportion and a relatively narrow common support region, but this does not necessarily violate the assumption—the key is whether $\hat{e}(L)$ overlaps between treated and control groups.

2.2 Estimation Principles

Under consistency, SUTVA, $Y(a) \perp\!\!\!\perp A \mid L$, and positivity, average potential outcomes are identified as

$$\mathbb{E}[Y(a)] = \mathbb{E}[\mathbb{E}[Y \mid A = a, L]] = \mathbb{E}\left[\frac{I(A = a)Y}{\mathbb{P}(A = a \mid L)}\right],$$

the right-hand side corresponding to outcome regression (g-formula) and inverse probability weighting (IPW) [6]. Both estimate the same $\mathbb{E}[Y(a)]$ but use data differently: the former models Y then marginalizes over L ; the latter models A then reweights the sample.

The intuition of IPW is to construct a pseudo-population where A and L are independent [10]. Let $e(L) = \mathbb{P}(A = 1 \mid L)$; then

$$\mathbb{E}[Y(a)] = \mathbb{E}\left[\frac{I(A = a)Y}{\mathbb{P}(A = a \mid L)}\right]$$

In finite samples, logistic regression estimates $\hat{e}(L) = \hat{\mathbb{P}}(A = 1 \mid L)$. For $A_i = 1$, stabilized weights are often $w_i = \hat{P}(A = 1)/\hat{e}(L_i)$; for $A_i = 0$, $w_i = (1 - \hat{P}(A = 1))/(1 - \hat{e}(L_i))$ to reduce weight variance [10]. ACE is the weighted prevalence difference between active and control groups. IPW requires only the propensity model $\hat{e}(L)$ to be correct (or close), not

the outcome model; but when $e(L)$ is near 0 or 1, weights become extreme and variance increases—hence the importance of positivity diagnostics [6, 10].

Parametric g-formula standardizes directly over L [6]:

$$\mathbb{E}[Y(a)] = \sum_l \mathbb{E}[Y \mid A = a, L = l] \mathbb{P}(L = l),$$

In practice, a logistic regression fits $\hat{\mathbb{P}}(Y = 1 \mid A, L)$, and the g-formula ACE is the sample mean difference in counterfactual predicted probabilities at $A = 1$ versus $A = 0$. G-formula does not rely on propensity scores, but misspecification of $\mathbb{E}[Y \mid A, L]$ (e.g., ignoring nonlinearity or interactions) still induces bias. When IPW and g-formula both have correct models under positivity, they converge to the same ACE; substantial disagreement often indicates severe misspecification of at least one model [9].

Doubly robust augmented IPW (AIPW) combines outcome regression and IPW [6, 14]. Let $\mu(a, L) = \mathbb{E}[Y \mid A = a, L]$; the individual-level estimator is

$$\hat{\psi}(a) = \hat{\mu}(a, L) + \frac{I(A = a)(Y - \hat{\mu}(a, L))}{\hat{e}(L)}, \quad \widehat{\text{ACE}} = \overline{\hat{\psi}(1)} - \overline{\hat{\psi}(0)}.$$

Doubly robustness means: if $\hat{e}(L)$ is correct, the IPW residual term corrects errors in $\hat{\mu}$; if $\hat{\mu}(a, L)$ is correct, then even if $\hat{e}$ is wrong, $\mathbb{E}[\hat{\psi}(a)]$ still equals $\mathbb{E}[Y(a)]$ [9, 14]. When a single model may misspecify $e(L)$ or $\mu(a, L)$, AIPW provides additional assurance—motivating my joint reporting of IPW, g-formula, and AIPW.

The propensity score $\pi(L) = \mathbb{P}(A = 1 \mid L)$ compresses multidimensional L to one dimension [13]. Rosenbaum and Rubin show: if $Y(a) \perp\!\!\!\perp A \mid L$, then $Y(a) \perp\!\!\!\perp A \mid \pi(L)$. Matching compares Y between treated and control with similar $\hat{\pi}(L)$; stratification divides the sample into propensity quantile groups, estimates effects within groups, then combines with weights [10, 13]. Like IPW, both aim to block backdoor paths but implement differently: matching/stratification emphasize local comparability; IPW emphasizes global reweighting. I use 1:1 nearest-neighbor matching (caliper = $0.2 \times \text{SD}(\hat{\pi})$): each active individual is matched to the nearest inactive control without replacement on the treated side, while controls may be reused; 54,939 active individuals are matched within the caliper (99.99% of the active group), with post-matching $|\text{SMD}| < 0.02$ on all seven covariates. Quintile PS stratification tests whether ACE depends on a specific weighting scheme [13]. Linear probability models (LPM) regress Y on A controlling L ; when $\mu(1, L) - \mu(0, L)$ is linear and exchangeability holds, LPM also identifies ACE, often close to g-formula, but lacks AIPW’s double robustness [6].

Model forms and inference details. The propensity score uses logistic regression with linear main effects for continuous covariates (age, BMI) and categorical covariates, without squared terms or interactions. The g-formula outcome model is logistic; AIPW combines logistic $\hat{e}(L)$ with an OLS (LPM) outcome model $\hat{\mu}(a, L)$. Standard errors: 500 nonparametric bootstrap replications for IPW and adjustment sensitivity; 400 and 300 bootstrap replications for AIPW and g-formula; HC1 heteroskedasticity-robust SEs for LPM; variance formulas for matched pairs and combined across-stratum variance for PS stratification. Bootstrap intervals treat the propensity score as fixed and do not propagate PS estimation uncertainty—a conventional simplification at the course-project level.

In summary, Section 3 reports crude prevalence difference, IPW, g-formula, AIPW, LPM, PS matching, and PS stratification—all targeting $\text{ACE} = \mathbb{E}[Y(1) - Y(0)]$ (prevalence difference) under identification assumptions. Convergence across methods supports

a narrative of consistent bias direction and adequate adjustment; divergence suggests model or positivity problems [6, 9].

3 Total Effect Estimation and Interpretation

Section 2 shows that under identification assumptions, ACE can be estimated via IPW, g-formula, AIPW, and PS methods; this section is the core empirical component—first reporting crude and adjusted ACE, then testing convergence across seven methods. If they converge, effect direction and magnitude are insensitive to estimation strategy; Section 4 further asks whether this convergence rests on testable conditions such as positivity.

3.1 Crude Comparison and the Need for Adjustment

Before adjustment, the inactive group ($A = 0$, $n = 13,446$) has CVD prevalence 53.3%; the active group ($n = 54,946$) has 48.5%. The crude prevalence difference is

$$\text{ACE}_{\text{crude}} = \bar{Y}_{A=1} - \bar{Y}_{A=0} = -0.0476 \quad [-0.0570, -0.0382],$$

with 95% CI excluding zero—the associational difference is statistically significant. Lower CVD rate among active individuals aligns directionally with protective associations summarized by Garcia et al. [2] and Mi et al. [1]; but crude comparison mixes two non-exchangeable sources: backdoor bias from L , and selective entry into activity.

Pre-weighting covariate differences (full results in Section 5, Table 2) show the active group has SMD of +0.065 for smoking and +0.064 for alcohol, slightly lower BMI (SMD = −0.035), slightly lower age (SMD = −0.026), and slightly higher cholesterol (SMD = +0.022). Smoking and alcohol increase CVD prevalence yet are more prevalent among active individuals—without controlling L , active-group CVD prevalence is artificially elevated, so this mechanism, in isolation, pulls the crude effect toward zero (underestimating true protection). Slightly lower BMI works in the opposite direction, pulling crude further from zero (overestimating true protection), partially offsetting this bias. Net effect: crude $|\text{ACE}|$ is slightly larger than adjusted, consistent with qualitative predictions from the backdoor structure, showing crude estimates cannot serve as causal conclusions and adjustment is necessary.

3.2 ACE Estimates from Seven Methods

After controlling L , Table 1 and Figure 2 summarize seven ACE estimates. Because the outcome is cross-sectional CVD prevalence, ACE below should be read as a prevalence difference—the effect of intervention on population disease proportion—not an incidence risk difference from cohort studies. By estimation principle: weighting/regression methods (IPW, AIPW, g-formula, LPM) yield −0.0439 to −0.0441; PS matching −0.0435 (SE= 0.0030, narrowest); PS stratification −0.0466 (SE= 0.0077, widest). Point estimates fall within a 3 percentage-point band [−0.047, −0.044], and all 95% CIs exclude zero—the primary empirical finding: different uses of $\hat{e}(L)$ and $\mathbb{E}[Y \mid A, L]$ do not produce divergent conclusions, and the protective effect is statistically robust.

Table 1: ACE of physical activity on CVD ($N = 68,392$)

Method	ACE	95% CI
Crude prevalence difference	−0.0476	[−0.0570, −0.0382]
IPW	−0.0440	[−0.0527, −0.0354]
Doubly robust AIPW	−0.0439	[−0.0520, −0.0358]
G-formula	−0.0439	[−0.0530, −0.0349]
LPM regression	−0.0441	[−0.0530, −0.0352]
PS matching	−0.0435	[−0.0494, −0.0376]
PS stratification	−0.0466	[−0.0616, −0.0315]

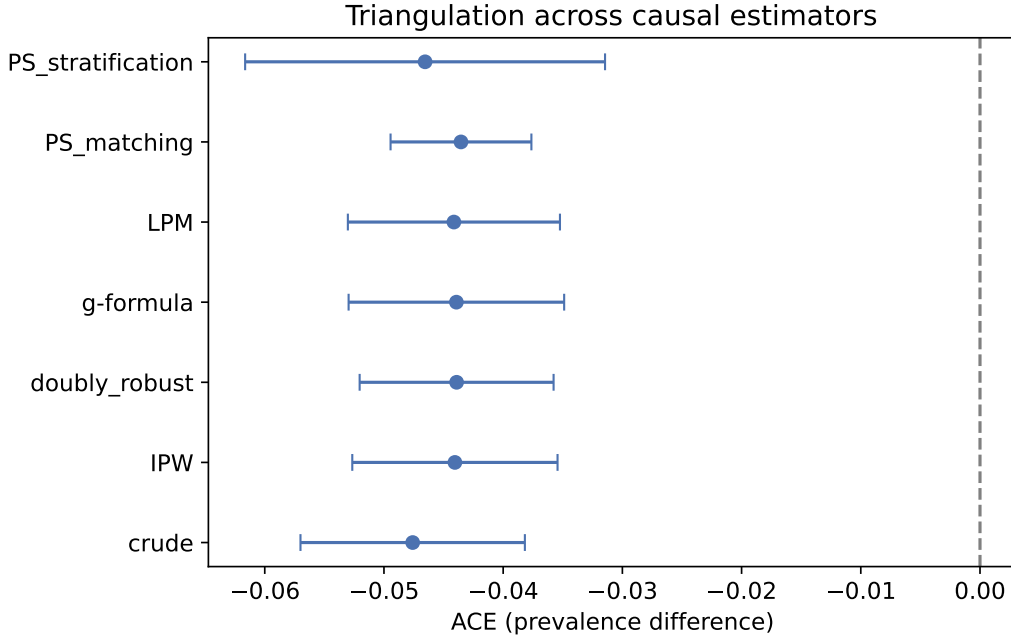


Figure 2: Forest plot of seven ACE estimates (horizontal axis: prevalence difference; vertical lines: 95% CIs; negative values indicate activity lowers CVD prevalence)

Using the IPW point estimate with inactive-group prevalence $p_0 = 0.533$ as reference, active-group prevalence is approximately $p_0 + \text{ACE} \approx 0.489$, giving approximate prevalence ratio (PR) ≈ 0.92 —protective direction consistent with literature, but because the outcome is CVD prevalence rather than mortality and activity is only binary self-report, magnitudes should not be numerically compared directly to Garcia et al.’s [2] CVD death RR (0.71). For this dataset, moving 80.3% of active individuals from the inactive counterfactual state to active would reduce population-average CVD prevalence by about 4.4 percentage points. The number needed to treat (NNT) is approximately $1/0.044 \approx 23$: roughly 23 inactive individuals would need to become active to prevent one additional case of prevalent CVD (an approximate interpretation on the cross-sectional prevalence scale), with clear public health interpretability.

This prevalence difference comprises two components: the genuine protective effect of PA on CVD development and progression, and reverse causation bias (individuals with existing CVD reducing activity). The estimate should not be equated with a pure

causal effect of PA on CVD incidence. Compared with Garcia et al.’s [2] prospective meta-analysis, the protective direction is fully consistent but the magnitude is smaller—a pattern with clear methodological rationale: (i) cross-sectional prevalence difference is subject to disease duration and survivor effects, which dilute the observed effect relative to incidence risk differences from cohort designs; (ii) reverse causation (CVD $\rightarrow$ reduced activity), by contrast, inflates the observed protective estimate (pulls ACE away from zero)—this works counter to (i) and (iii); (iii) binary self-reported activity without intensity or dose information introduces non-differential misclassification, biasing effect estimates toward the null. The net reduction relative to incidence-based meta-analyses is therefore driven mainly by the prevalence versus incidence scale difference and measurement error.

3.3 Why Multiple Methods Converge

Why do IPW (relying on $\hat{e}$), g-formula (relying on $\hat{\mu}$), AIPW (combining both), and PS matching/stratification (relying on local comparability via $\hat{\pi}$) yield nearly identical ACE? From an identification perspective, all target $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$. Here, IPW and g-formula differ by only 0.0001 (-0.0440 vs -0.0439), and AIPW nearly coincides with both—meaning propensity and outcome models do not conflict enough at the ACE level to split conclusions; severe misspecification of one typically separates IPW and g-formula [9]. AIPW not triggering single-model compensation means both working models mutually support—the core value of triangulation: not repeating one regression, but cross-validating the same estimand via different identification paths [6, 9].

Method by method: IPW and g-formula correspond to the two formulas in Section 2.2—the former reweights so $A \perp\!\!\!\perp L$, the latter marginalizes $\mathbb{E}[Y | A, L]$; equivalence under correct models is empirically reflected here. AIPW is consistent if either $\hat{e}$ or $\hat{\mu}$ is correct in the Robins sense [14]; coincidence with IPW indicates at least not the worst case of both models failing. LPM coefficient -0.0441 nearly equals g-formula, as expected under linear $\mathbb{E}[Y | A, L]$ [6].

PS matching (-0.0435) is slightly smaller in magnitude than IPW (-0.0440) with narrower SE (0.0030 vs 0.0044), consistent with “local randomization” among similar $\hat{\pi}$ and exclusion of non-comparable individuals outside the caliper—the matched sample is more homogeneous, estimates are more precise, but may slightly deviate from the full-sample ACE. PS stratification (-0.0466) has the point estimate closest to crude and the widest interval, because $A = 0$ individuals are sparse within quintile strata and within-stratum variance is large—directly related to narrow positivity under high activity prevalence (Section 4). Nevertheless, stratification still falls within other methods’ CIs, showing the effect is not an artifact of one weighting scheme.

The difference between crude and adjusted ACE (-0.0476 vs -0.0440) is also understandable from the backdoor structure: after controlling L , smoking/alcohol’s upward pull on active-group CVD prevalence is removed, and ACE absolute value slightly decreases, consistent with Section 3.1 confounding direction. In sum, multi-method triangulation supports: under control of L and identification assumptions, activity reduces average CVD prevalence by about 4.4 percentage points. But convergence alone does not prove positivity or exchangeability—Section 4 turns to positivity, the only assumption in Section 2.1 that is testable from data.

4 Positivity Assessment: Propensity Score Overlap

Section 3 ACE estimates rely on IPW and PS methods, both requiring positivity: $0 < e(L) < 1$ on the support of L [6, 10]. If treated and control $\hat{e}(L)$ barely overlap, then $\mathbb{E}[Y(0)]$ is unidentified in regions lacking controls, and Section 3 numerical agreement may rest on extrapolation rather than identification. This section tests common support via propensity score distributions.

4.1 Common Support and Distributional Overlap

Let $\hat{e}(L) = \hat{\mathbb{P}}(A = 1 \mid L)$. Figure 3 shows densities of $\hat{e}$ for $A = 0$ and $A = 1$. With 80.3% active, the logistic propensity model assigns high $\hat{e}$ to most individuals: inactive group ($n = 13,446$) mean $\hat{e}$ 0.802, range $[0.742, 0.874]$; active group ($n = 54,946$) mean 0.804, range $[0.745, 0.879]$. Common support is $\hat{e} \in [0.745, 0.874]$; 99.99% of individuals fall in this interval. Positivity is not empirically violated—at each $\hat{e}$ level, both active and inactive individuals exist, IPW weights do not systematically explode from complete lack of controls, and PS matching finds controls within the caliper.

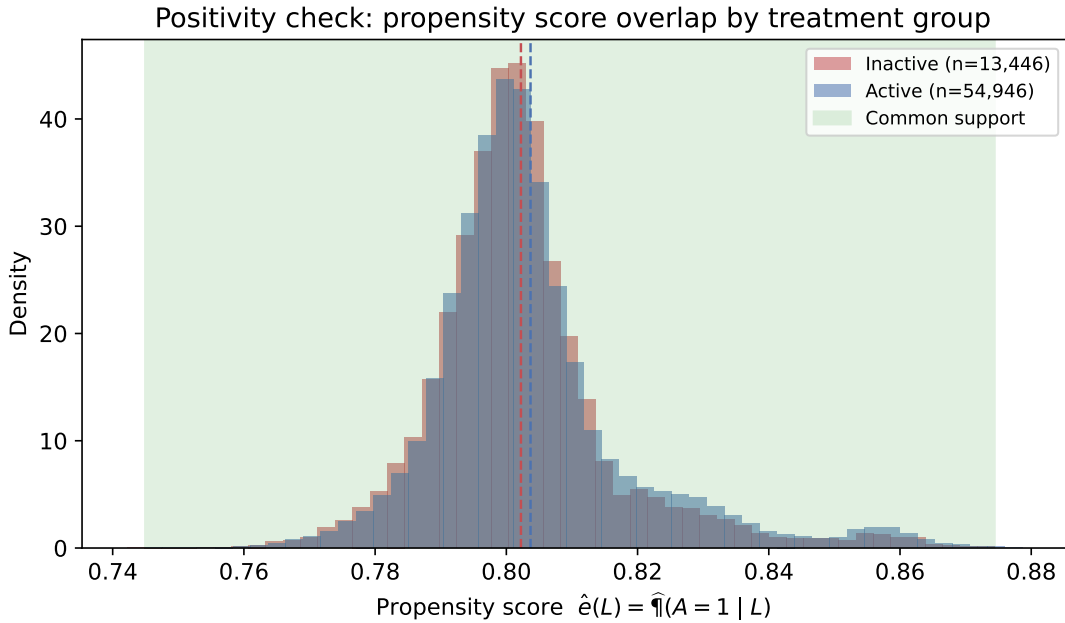


Figure 3: Positivity assessment: propensity score distributions by treatment group and common support region (green). Dashed lines indicate group PS means.

From weight structure: for inactive individuals, stabilized IPW weight is $w_i = (1 - \hat{P}(A = 1)) / (1 - \hat{e}(L_i))$ with $\hat{P}(A = 1) = 0.803$. At mean $\hat{e} = 0.802$, $w_i \approx 0.197/0.198 \approx 1.0$; for active individuals, $w_i = \hat{P}(A = 1) / \hat{e}(L_i) \approx 0.803/0.802 \approx 1.0$. Even at upper common-support bound $\hat{e} = 0.874$, inactive weight is about $0.197/0.126 \approx 1.56$, moderate rather than explosive. Across the full sample, stabilized weights have median 1.00, 95th percentile 1.04, and maximum 1.57; the fraction with weight > 5 is 0. Section 3 IPW is not dominated by extreme weights—positivity is not only formally satisfied but corresponds to acceptable weight distributions [10].

Positivity assessment answers whether data contain control information needed to compare $\mathbb{E}[Y(1)]$ and $\mathbb{E}[Y(0)]$, not whether exchangeability holds. Overlap plots at least exclude the most fatal identification failure—complete lack of overlap [10]—providing a necessary (not sufficient) condition for Section 3 IPW/PS estimates.

4.2 Identification Implications Under High Activity Prevalence

Structural features of this data must also be noted: group mean $\hat{e}$ values are nearly identical (0.802 vs 0.804), common support width only about 0.13—almost everyone is predicted “high activity propensity.” ACE identification therefore relies mainly on conditional probability variation within L rather than broad dispersion of $\hat{e}$ near 0. If L ’s explanatory power for A increases further and $\hat{e}$ concentrates toward 1, positivity will tighten marginally [10].

This pattern is consistent with high activity prevalence (80.3%): marginally only 19.7% are inactive, but conditionally individuals with $e(L) < 1$ may still exist—Figure 3 tests this directly. Quintile strata each contain about 13,678–13,679 individuals, with 79–83% active and 17–21% inactive controls (only 17.0% inactive in the fifth stratum); each stratum retains enough controls for within-stratum prevalence differences, but the fifth stratum has the fewest inactive individuals, consistent with the widest PS stratification CI (SE= 0.0077). Global IPW uses all 13,446 inactive individuals as controls, with SE (0.0044) much smaller than PS stratification. If positivity truly failed (e.g., $\hat{e} \rightarrow 1$ with no $A = 0$ individuals), I would simultaneously see overlap disappearance, weight explosion, and PS matching failure to pair—patterns not observed here.

The relationship between positivity and Section 3 conclusions: multi-method convergence in Section 3 is obtained under the premise that IPW/PS identification is feasible within common support; this section excludes the most severe case where estimates rely purely on extrapolation from lack of controls. The next section further tests whether IPW truly balances L within common support (Section 5, Table 2, shows post-weighting $|\text{SMD}| < 0.003$), verifying backdoor blocking in data.

5 Covariate Balance and IPW Validity

Section 4 shows positivity holds in data, giving IPW/PS estimates the necessary conditions for identification. But positivity only guarantees controls exist, not that IPW weights eliminate distributional differences in L between treated and control—if L remains imbalanced after weighting, backdoor blocking implied by $Y(a) \perp\!\!\!\perp A \mid L$ fails at the data level and Section 3 ACE may still suffer residual confounding [10, 13]. This section quantifies covariate balance before and after IPW via love plots and SMD, testing whether Section 2.2 IPW “pseudo-randomization” is achieved on measured L .

5.1 Balance Principle of IPW Weighting

Rosenbaum and Rubin’s [13] core result: if $Y(a) \perp\!\!\!\perp A \mid L$, then $Y(a) \perp\!\!\!\perp A \mid \pi(L)$ where $\pi(L) = \mathbb{P}(A = 1 \mid L)$. Further, within strata of similar propensity scores, L distributions should approximate balance between treated and control—the testable implication of IPW validity [10]. With stabilized weights w_i as in Section 2.2, post-weighting stan-

standardized mean difference (SMD) for covariate j is

$$\text{SMD}_j = \frac{\bar{L}_{j,A=1}^w - \bar{L}_{j,A=0}^w}{s_j},$$

where $\bar{L}^w$ is the weighted mean and s_j the pooled standard deviation. Empirically $|\text{SMD}| < 0.1$ is acceptable balance [10]; post-weighting $|\text{SMD}| > 0.1$ suggests omitted covariates in the propensity model, positivity only in a narrow region, or insufficient IPW weighting [6, 10].

Balance diagnostics test whether backdoor paths on measured L are blocked, not whether exchangeability holds. With unobserved U , even perfect balance on L may leave $Y(a) \not\perp\!\!\!\perp A \mid L$ violated—Section 7 E-value quantifies how strong U would need to be to explain away Section 3 estimates [12].

5.2 Covariate Differences Before and After Weighting

Table 2 and Figure 4 report SMD for seven adjustment covariates. Before weighting, active and inactive groups differ systematically: smoking ($|\text{SMD}| = 0.065$) and alcohol (0.064) show the largest differences, with higher prevalence among active individuals—consistent with Section 3.1 confounding direction; BMI (0.035), age (0.026), cholesterol (0.022), glucose (0.020) also differ visibly; sex (0.014) is smallest but nonzero. These differences explain why crude and adjusted ACE differ: without weighting, backdoor paths $A \leftarrow L \rightarrow Y$ are not blocked [6].

After IPW weighting, all seven covariates have $|\text{SMD}| < 0.003$ (gender 0.001, smoke/alco < 0.001 , BMI 0.003, gluc 0.002, age/cholesterol < 0.001). Maximum imbalance compresses from 0.065 to under 0.003, a reduction exceeding 95%. In the pseudo-population, A and L are approximately independent, showing the logistic propensity model $\hat{e}(L)$ successfully achieves post-weighting covariate balance—direct evidence that IPW operates effectively as a backdoor adjustment tool [10].

Table 2: SMD for all adjustment covariates before and after IPW weighting

Variable	Pre-weight SMD	Post-weight SMD
Smoking (smoke)	+0.065	−0.000
Alcohol (alco)	+0.064	−0.000
BMI	−0.035	−0.003
Age	−0.026	−0.000
Cholesterol	+0.022	+0.000
Glucose (gluc)	−0.020	−0.002
Sex (gender)	+0.014	+0.001

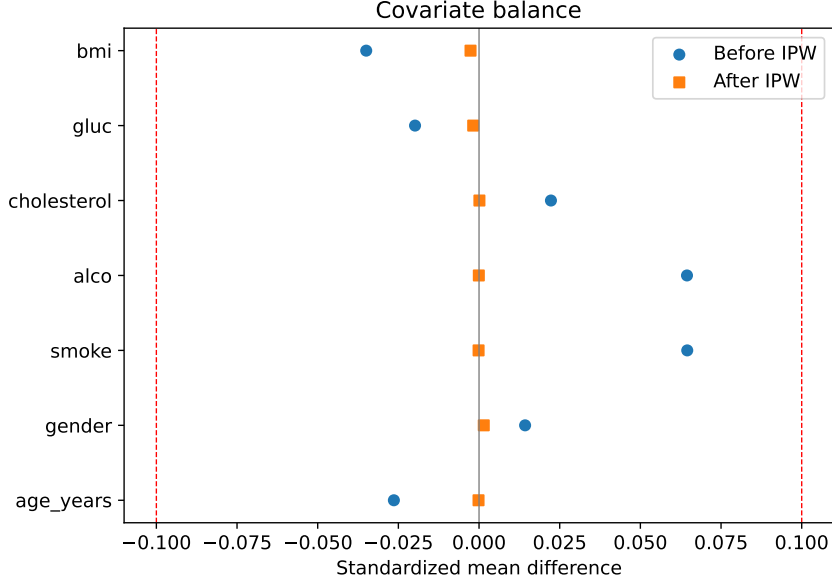


Figure 4: Covariate SMD before and after IPW weighting. Red dashed lines indicate ± 0.1 thresholds.

5.3 Implications of Balance Diagnostics for Section 3 ACE

Linking Sections 4–5 diagnostics with Section 3 conclusions forms a complete testable evidence chain: Section 4 positivity shows common support holds (99.99% in $\hat{e} \in [0.745, 0.874]$), IPW weights do not explode (inactive weight ≈ 1.56 at upper bound); this section’s love plot shows good post-weighting balance on L . Together they support that Section 3 IPW ACE = -0.0440 is not a spurious estimate under lack of controls or severe post-weighting imbalance [6, 10].

Good balance does not prove U is absent. Smoking/alcohol—“negative health behaviors” slightly more common among active individuals—are balanced by IPW; but unmeasured U (family history, diet) correlated with both activity and CVD may still violate exchangeability. Positivity and balance exclude the most severe identification failures at the testable level; Section 6 examines mechanisms given stable total effects, and Section 7 further bounds conclusion strength via adjustment specifications and E-value [12].

6 Mechanism Analysis: BMI Mediation Path

Sections 3–5 establish that after controlling L , total effect ACE ≈ -0.044 , with IPW identification conditions verified in data. The next question: how much of this protective effect passes through BMI? Chen and Wang et al. report PA mediation through obesity-related indices among Chinese adults and in recreational activity–hypertension studies [3, 4]; Hall et al. [5] note more complex mechanisms when considering CRF. This section, under VanderWeele’s regression-based mediation framework [11], separates BMI from L as mediator M , decomposing NDE and NIE, with comparison to concurrent blood pressure mediation and sensitivity analysis.

6.1 Mediation Estimands and Identification Framework

In Figure 1, the total-effect model includes BMI in L to block backdoor path $A \leftarrow M \rightarrow Y$; mechanism analysis requires a different estimand: controlling $L \setminus \{M\} = \{\text{age, sex, smoking, alcohol, cholesterol, glucose}\}$, define

$$\text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad \text{NIE} = \mathbb{E}[Y(0, M(1)) - Y(0, M(0))],$$

with total effect $\text{TE} = \text{NDE} + \text{NIE}$ [11]. NDE is the effect of activity on CVD when the $A \rightarrow M$ path is blocked; NIE is the effect transmitted only through BMI change. Under linear LPM approximation, decomposition uses a - and b -paths:

$$\text{NIE} = a \times b, \quad a = \mathbb{E}[M(1) - M(0) \mid L \setminus M], \quad b = \mathbb{E}[Y(1, m) - Y(0, m) \mid L \setminus M, M = m].$$

Identification requires additional assumptions [11]: (i) no unmeasured confounders between M and Y (no $M \leftarrow U' \rightarrow Y$); (ii) $L \setminus M$ sufficiently controls confounding for $A \rightarrow M$ and $A \rightarrow Y$; (iii) **no exposure–mediator interaction** (by default no $A \times M$ interaction in the linear framework; if interaction exists, NDE/NIE decomposition shifts slightly); (iv) cross-world independence. In this study, the BMI–CVD association may still be confounded by diet, lifestyle, and other U' , so NIE and the 5.6% mediated share are **less credible than the total effect** and should be read as exploratory. A single cross-sectional BMI measurement cannot capture dynamic weight change—limitations to remember when interpreting NIE proportion.

6.2 BMI Path Decomposition Results

Table 3 summarizes decomposition (500 bootstrap CIs). a -path coefficient -0.183 : active individuals have mean BMI 0.18 units lower (approximately 0.18 kg/m²), confirming activity associates with lower BMI. b -path $+0.014$: controlling $L \setminus M$, each 1-unit BMI increase raises CVD prevalence by 1.4 percentage points. Product $\text{NIE} = -0.183 \times 0.014 \approx -0.0026$ $[-0.0040, -0.0013]$: activity indirectly lowers CVD prevalence through BMI by about 0.26 percentage points. $\text{NDE} = -0.0441$ $[-0.0530, -0.0362]$, about 94% of total effect $\text{TE} = -0.0468$ $[-0.0562, -0.0383]$; indirect path about 5.6%.

Table 3: BMI mediation path decomposition (controlling $L \setminus M$: age, sex, smoking, alcohol, cholesterol, glucose)

Effect / path	Estimate	95% CI
a -path ($A \rightarrow M$)	-0.183	—
b -path ($M \rightarrow Y \mid A$)	$+0.014$	—
NIE (indirect effect)	-0.0026	$[-0.0040, -0.0013]$
NDE (direct effect)	-0.0441	$[-0.0530, -0.0362]$
Total effect TE	-0.0468	$[-0.0562, -0.0383]$
Indirect effect proportion	5.6%	—

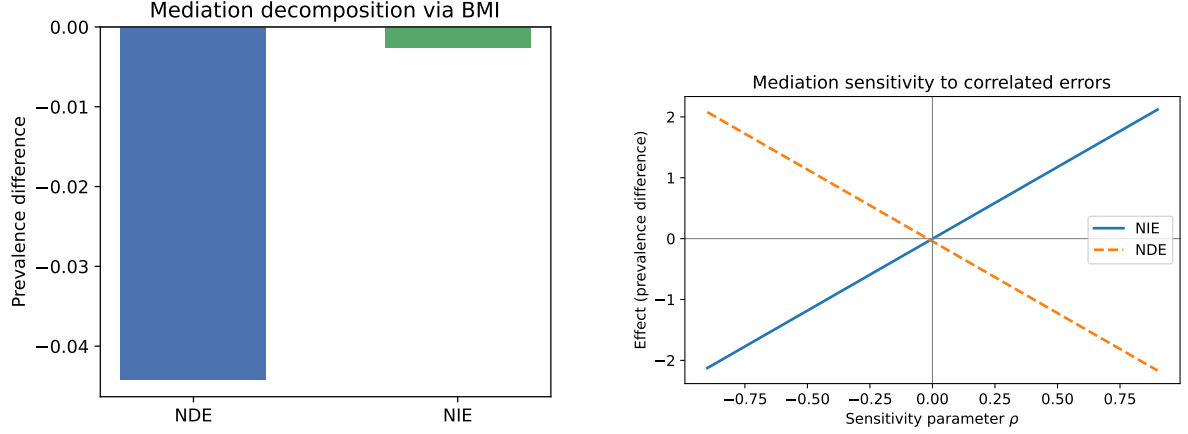


Figure 5: Left: NDE and NIE decomposition; right: mediation sensitivity—NIE change as ρ (correlation of mediator/outcome model residuals) varies

Figure 5 (left) visually shows NDE dominance. Compared with Section 3: IPW total effect excluding BMI as confounder is -0.0468 (Section 7), matching mediation TE = -0.0468 —because BMI is no longer controlled as confounder in the mechanism model, total effect is slightly larger than main-analysis ACE (-0.0440), consistent with DAG expectations when BMI is both confounder and mediator [11]. NDE = -0.0441 nearly equals main-analysis LPM/IPW, meaning Section 3 total effect is essentially close to the direct effect not passing through BMI; the BMI channel contributes only about 5.6% additional protection.

6.3 Comparison with Literature and Blood Pressure Mediation

Does NDE at 94% contradict literature? Not necessarily. Chen et al. [3] report PA mediation through body fat, blood pressure, lipids, and other pathways among Chinese adults—I test only the BMI single path, with cross-sectional binary activity; the a -path is significant but limited in magnitude. Biologically, activity may also affect CVD through endothelial function, systemic inflammation, cardiorespiratory fitness, etc., not necessarily via BMI [5, 15]. Wang et al. [4] find obesity indices mediate PA–hypertension associations, but the outcome here is CVD not hypertension, and BMI is only a coarse obesity proxy.

As design contrast, replacing mediator with concurrent systolic pressure *ap_hi* yields NIE = $+0.0006$, NDE = -0.0447 , total effect = -0.0441 —indirect path nearly zero. Cross-sectional blood pressure reflects two opposing pathways: a protective path “activity→lower blood pressure→lower CVD” and a pathological path “existing CVD→elevated blood pressure”; on a single examination these partially cancel, yielding near-zero observed indirect effect. This does not mean blood pressure is unimportant as a mediator in the real world, but that cross-sectional design cannot identify the dynamic chain activity→blood pressure change→CVD [6, 15]. This does not contradict significant blood pressure mediation in longitudinal cohorts, but highlights the value of longitudinal data for mechanism identification.

6.4 Mediation Sensitivity Analysis

Figure 5 (right) reports VanderWeele-style sensitivity [11]: if unmeasured factors correlate mediator and outcome model residuals (parameter ρ), NIE rises with ρ . Here, mediator/outcome residual SDs are 5.02 and 0.47; at $\rho \approx 0.001$, NIE is fully offset ($\rho_{\text{tip}} \approx 0.0011$). Because NIE itself is only 5.6%, small ρ can flip the indirect effect—mainly threatening mechanism interpretation (“does activity work through BMI?”) rather than total effect: scanning ρ , NDE remains stable around -0.044 , consistent with Section 3 IPW ACE. Thus, the conclusion that activity lowers CVD does not depend on BMI mediation; even if NIE is fully explained by unmeasured factors, NDE still supports a protective total effect.

7 Heterogeneity, Adjustment Specifications, and Unmeasured Confounding

Section 3 gives overall ACE; Sections 5–6 show identification diagnostics pass and BMI mediation is a small share. This section stress-tests conclusions from three complementary angles: consistency across subgroups (age, sex, BMI tertiles, and smoking), dependence on adjustment specifications, and strength of unmeasured confounding needed to explain away estimates (E-value). Together they bound extrapolation of Section 3 causal conclusions [6, 12].

7.1 Age-Stratified Heterogeneity

Repeating IPW by age quartiles (Q1 youngest $\rightarrow$ Q4 oldest), Figure 6 (left) reports stratum ACE. Q1: -0.041 $[-0.059, -0.023]$; Q2: -0.039 $[-0.057, -0.022]$; Q3: -0.047 $[-0.066, -0.029]$; Q4: -0.050 $[-0.068, -0.031]$. All four 95% CIs exclude zero and overlap substantially—protective effects exist across age groups without statistically significant effect modification.

Age-stratified IPW re-estimates $\hat{e}(L)$ and ACE within each stratum, testing whether protection is driven by one age group. Q4 point estimate is slightly larger (-0.050 vs Q2 -0.039), possibly because older groups have higher baseline CVD prevalence and greater absolute prevalence reduction space from activity [1]; but within-stratum sample sizes are smaller than full sample, SE about 0.009 (full-sample IPW SE 0.004), CIs are wider, insufficient to assert significant heterogeneity. This supports that full-sample Section 3 ACE is not an artifact of one age stratum but a protective direction across the life course.

7.2 Exploratory Heterogeneity by Sex and BMI

Beyond age, I repeat IPW by sex and BMI tertiles (excluding sex from the adjustment set within sex strata) to add public-health-relevant dimensions. Women ($n = 44,550$) have IPW ACE -0.035 ; men ($n = 23,842$) have -0.059 —men show a larger point estimate in absolute value, but this is exploratory stratification without a formal sex interaction test, so I do not claim activity is more effective for men. Low/middle/high BMI tertiles yield IPW ACE of -0.046 , -0.043 , and -0.044 , respectively—consistent direction and similar magnitude, echoing Section 6’s small BMI-mediated share. These results describe **stratum-specific effects**, not causal effect modification identified via interaction terms.

7.3 Effect Modification by Smoking

Adding activity $\times$ smoking interaction in LPM (controlling remaining $L \setminus \{\text{smoke}\}$), Figure 6 (right) reports stratum effects. Non-smokers: ACE -0.0417 $[-0.0510, -0.0324]$; smokers: -0.0734 $[-0.1079, -0.0389]$; interaction coefficient -0.032 , $p = 0.062$, marginally significant.

Smokers have higher baseline CVD prevalence (smoking is a strong risk factor in L), so larger absolute prevalence reduction from activity in that subgroup is not surprising [16]. But smoker CI is wide (upper bound -0.039 still significantly negative, lower bound -0.108 indicates large uncertainty), and $p = 0.062$ does not reach conventional 0.05—I treat this as an exploratory finding, not asserting activity is more effective for smokers. Importantly, among non-smokers alone, ACE remains -0.042 and significant; Section 3 conclusions do not depend on smoking interaction.

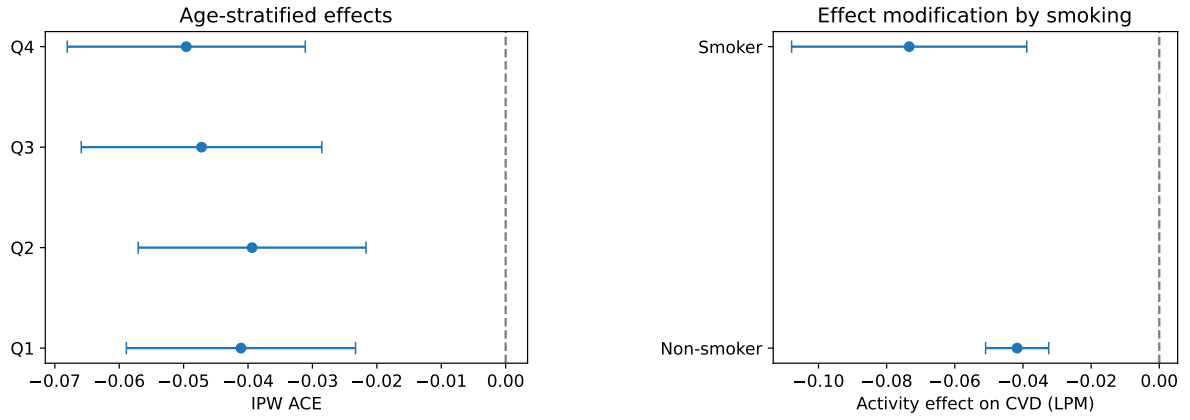


Figure 6: Left: IPW ACE by age quartile with 95% CIs; right: LPM effects stratified by smoking with 95% CIs

7.4 Adjustment Set Specification Sensitivity

Adjustment set choice directly changes estimand meaning [6, 11]. Figure 7 compares three IPW specifications (Table 4): main analysis (with BMI) ACE = -0.0440 $[-0.0534, -0.0347]$; excluding BMI -0.0468 $[-0.0557, -0.0379]$; additionally including systolic/diastolic pressure -0.0446 $[-0.0530, -0.0361]$.

Table 4: IPW ACE under different adjustment sets

Specification	ACE	95% CI
Main analysis (L includes BMI)	-0.0440	$[-0.0534, -0.0347]$
Exclude BMI	-0.0468	$[-0.0557, -0.0379]$
Add blood pressure (ap_hi , ap_lo)	-0.0446	$[-0.0530, -0.0361]$

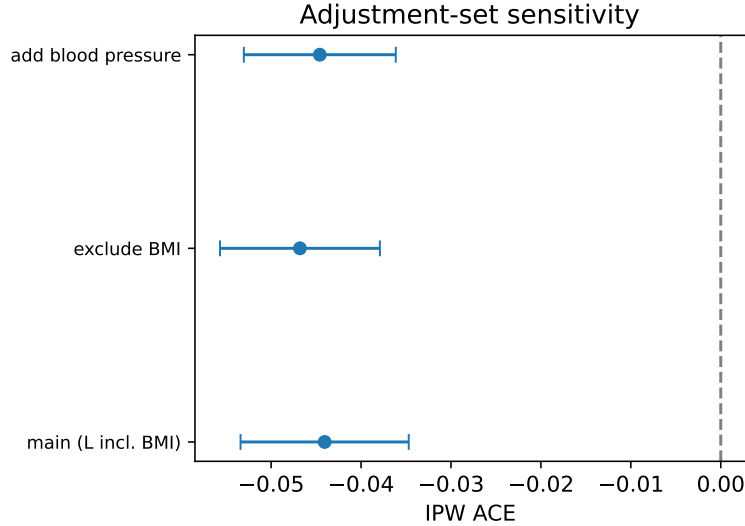


Figure 7: IPW ACE and 95% CIs under different adjustment sets

Excluding BMI increases ACE magnitude by about 0.003, consistent with Section 6 mechanism analysis: BMI is both confounder on $A \rightarrow Y$ and mediator on $A \rightarrow M \rightarrow Y$; main analysis controlling BMI estimates total effect including BMI confounding paths; excluding BMI partially opens mediation paths, shifting TE toward -0.0468 [11]. Adding blood pressure barely changes ACE (-0.0446 vs -0.0440), showing concurrent blood pressure does not strongly drive conclusions—neither supporting “must control blood pressure” nor extreme “effect greatly attenuated after controlling blood pressure,” consistent with Section 1 design choice not to control blood pressure in main analysis [6]. All three specifications’ CIs overlap substantially; Section 3 ACE is robust to reasonable adjustment changes.

7.5 E-Value and Unmeasured Confounding

Exchangeability $Y(a) \perp\!\!\!\perp A \mid L$ is untestable; E-value quantifies how strong unmeasured confounding U would need to be to fully explain away the observed effect [12]. Using IPW point estimate $ACE = -0.0440$ and inactive-group CVD prevalence $p_0 = 0.533$, active-group prevalence is approximately $p_1 \approx 0.489$, giving approximate prevalence ratio (PR) ≈ 0.92 (note: E-value is computed based on the prevalence ratio; its causal interpretation is inherently weaker than E-values from incidence risk ratios in cohort designs). E-value point estimate is about 1.40; for the lower bound of the 95% CI on ACE, E-value is about 1.46—meaning if unmeasured U exists, its associations with both activity and CVD (on the ratio scale) would each need to reach about 1.4 to fully offset the observed protective effect [12].

An E-value of 1.40 is not high: in epidemiology, smoking–CVD associations often reach $RR \approx 1.8$ – 2.0 , and family history $RR \approx 1.5$ – 2.0 [1, 6]—unmeasured confounding of family-history strength could fully offset the observed effect on the prevalence-ratio scale. Family history, diet, socioeconomic status, and other U may plausibly reach this magnitude of association with both activity and CVD. Because the outcome is cross-sectional CVD prevalence, the E-value here reflects unmeasured confounding strength on prevalence-scale associations; its causal force is weaker than E-values derived from

incidence risk ratios in prospective cohort designs. Thus, even with Section 3–5 multi-method convergence and passing positivity and balance, I position causal conclusions as cross-sectional supportive evidence consistent across methods and diagnostics, not RCT-level confirmation. E-value closes the loop with Section 5’s point that U may still threaten exchangeability: testable conditions are good, untestable assumptions require caution.

Linking Section 7 with the full paper: Section 3 ACE is stable; Sections 4–5 testable identification conditions hold; Section 6 mechanisms are NDE-dominated; this section shows effects are robust across age strata and BMI tertiles, exploratory sex differences warrant caution, adjustment specifications are insensitive, but E-value indicates residual confounding cannot be ignored—providing complete basis for Section 8 conclusions and limitations.

8 Conclusions and Limitations

8.1 Main Conclusions

In observational data, this study conducts a complete causal inference pipeline around whether physical activity causally lowers CVD prevalence: identification assumptions, multi-method estimation, testable diagnostics, mechanism and sensitivity analysis. Methodologically, under DAG and backdoor guidance I specify the average causal effect estimand, sequentially applying IPW, g-formula, doubly robust AIPW, LPM, and PS matching/stratification—principally different strategies—and testing positivity and covariate balance via propensity score overlap and post-weighting balance; I further decompose BMI mediation under VanderWeele’s framework, combining adjustment-set variation, subgroup analysis, and E-value to assess conclusion strength [6, 11, 12].

Main findings summarize as three points. First, after controlling age, sex, smoking, alcohol, and major metabolic indicators, multiple methods agree closely in direction and magnitude on the total effect (prevalence difference ≈ -0.044 ; NNT ≈ 23), all supporting a protective association between physical activity and CVD prevalence—showing conclusions do not depend on one weighting or regression specification but have reasonable methodological robustness under stated assumptions [9]. Second, positivity and post-IPW covariate balance diagnostics show testable identification conditions are met in data; Section 3 effect estimates do not rest on fragile foundations of lack of controls or severe post-weighting imbalance [10, 13]. Third, mechanism analysis shows protection operates mainly through direct paths not mediated by BMI, with small indirect share via BMI; effects are protective across age strata and BMI tertiles and insensitive to reasonable adjustment changes, but E-value analysis indicates unmeasured confounding may still weaken causal interpretation [12].

In practical terms, conclusions align with Garcia et al. [2], Mi et al. [1], and extensive observational evidence that insufficient physical activity is a modifiable CVD risk factor: population-level promotion of regular activity and reduced sedentary behavior retains clear public health value. Beyond a protective point estimate, this paper demonstrates on a single observational dataset how to combine causal estimand specification, multi-method triangulation, and testable diagnostics to make the inference from association to causation transparent and reproducible—a reference for future analyses of similar examination or cohort data.

8.2 Limitations and Outlook

Several limitations require attention in interpretation and generalization. Data are cross-sectional, all variables measured at one examination time, so **reverse causation** (individuals with CVD reducing activity) cannot be excluded—this bias direction would inflate the apparent protective effect (more negative ACE); the observed 4.4 percentage-point prevalence difference may partly reflect reverse causation rather than purely prospective causal effect. Activity is binary self-report without intensity, frequency, or objective measurement, not one-to-one with guideline MVPA dose; non-differential misclassification may also attenuate effects toward the null [1]. Exchangeability is fundamentally untestable; despite passing positivity and balance, unmeasured family history, diet, socioeconomic status, etc., may constitute residual confounding, and E-value analysis shows confounding strength needed to fully explain away observed effects is not extreme [6, 12]. Mediation analysis relies on linear approximation and cross-sectional BMI measurement; answers about biological pathways through which activity lowers CVD remain preliminary [11].

In sum, I position causal conclusions as cross-sectional supportive evidence consistent across methods and diagnostics, not randomized-trial confirmation. Future work may validate in longitudinal follow-up, natural experiments, or instrumental variable/MR designs; combining accelerometer activity measurement with richer covariates could improve estimand interpretability and extrapolation strength. All code, data, figures, and numerical output are public on GitHub (Appendix C) for reproduction and extension.

Appendix

A Figure and Table Index

Main tables in the text include: Table 1 (seven ACE estimates), Table 2 (covariate balance), Table 3 (BMI mediation decomposition), Table 4 (adjustment sensitivity). Main figures are listed below.

Table 5: Index of main figures in the text

Fig.	Filename	Content
1	(TikZ)	Causal directed acyclic graph
2	forest_plot.pdf	Forest plot of seven ACE estimates
3	ps_overlap.pdf	Propensity score overlap (positivity)
4	love_plot.pdf	SMD before and after IPW weighting
5	mediation_*.pdf	BMI mediation decomposition and ρ sensitivity
6	age_heterogeneity.pdf, etc.	Age stratification and smoking effect modification
7	adjustment_sensitivity.pdf	Adjustment set sensitivity

B Notation

Table 6: Summary of main notation used in the text

Symbol	Name	Meaning
N	Sample size	Analysis sample after cleaning ($N = 68,392$)
n	Subgroup size	Number of observations in a stratum or treatment arm
i	Individual index	Index for unit i
A, A_i	Treatment	Binary physical activity ($A = 1$ active, $A = 0$ inactive)
Y, Y_i	Outcome	CVD diagnosis indicator ($Y = 1$ if diseased)
$Y(a), Y_i(a)$	Potential outcome	Outcome under treatment level a
L	Covariate set	Adjustment set (age, sex, smoking, alcohol, cholesterol, glucose, BMI, etc.)
M	Mediator	BMI (separated from L in mechanism analysis)
U	Unmeasured factor	Unobserved confounding (e.g., family history, diet)
l	Covariate value	A realization of L
ACE	Average causal effect	$\mathbb{E}[Y(1) - Y(0)]$, measured as cross-sectional CVD prevalence difference
NDE	Natural direct effect	Effect of activity when the $A \rightarrow M$ path is blocked

Table 6 (continued)

Symbol	Name	Meaning
NIE	Natural indirect effect	Effect transmitted through M
TE	Total effect	NDE + NIE
SMD_j	Standardized mean difference	Balance metric for covariate j before/after weighting
RR	Risk ratio	Ratio of risks in treated vs. control groups
a	$A \rightarrow M$ path coefficient	Effect of activity on BMI in mediation decomposition
b	$M \rightarrow Y$ path coefficient	Effect of BMI on CVD given $L \setminus M$
$\mathbb{E}[\cdot]$	Expectation	Population or potential-outcome average
$\mathbb{P}(\cdot)$	Probability	Probability measure
$\perp\!\!\!\perp$	Conditional independence	Statistical independence given conditioning variables
$e(L)$	Propensity score	$\mathbb{P}(A = 1 \mid L)$
$\pi(L)$	Propensity score	Same as $e(L)$, i.e., $\mathbb{P}(A = 1 \mid L)$
$\hat{e}(L), \hat{\pi}(L)$	Estimated propensity score	$\hat{\mathbb{P}}(A = 1 \mid L)$ from logistic regression
$\mu(a, L)$	Outcome regression	$\mathbb{E}[Y \mid A = a, L]$
$\hat{\mu}(a, L)$	Estimated outcome model	Fitted $\mathbb{E}[Y \mid A, L]$
$\hat{\psi}(a)$	AIPW individual estimate	Individual-level contribution in doubly robust estimation
w_i	IPW weight	Stabilized inverse-probability weight
$\bar{L}^w$	Weighted mean	IPW-weighted sample mean of a covariate
s_j	Pooled SD	Denominator used in SMD calculation
p_0, p_1	Risk	CVD prevalence in inactive/active groups
ρ	Sensitivity parameter	Correlation of mediator/outcome model residuals
ρ_{tip}	Tipping point	Critical ρ at which NIE is nullified
ap_hi, ap_lo	Blood pressure	Systolic/diastolic pressure (robustness analysis)
$I(A = a)$	Indicator function	Equals 1 if $A = a$, else 0

C Code Repository and Reproduction Instructions

Complete project materials (dataset, analysis scripts, numerical output, figures, and LaTeX report source) are available at the public GitHub repository:

[GitHub: Liesen-Gai-Causal-Inference-Final-Project](#) .

C.1 Repository Structure

```
Liesen-Gai-Causal-Inference-Final-Project/
|-- README.md
|-- code/
|   '-- analysis.py          # main pipeline
|-- data/
|   '-- cardio_train.csv     # Ulianova (70,000 raw rows)
|-- output/
|   |-- summary.json
|   |-- report_metrics.json
|   |-- estimates.csv
```

```
|   |-- balance.csv
|   '-- ...
|-- figures/                # report figures (PDF)
'-- report/
    |-- final_pj_en.tex      # LaTeX source (English)
    |-- final_pj_en.pdf      # compiled report PDF
    '-- references.bib       # BibTeX bibliography
```

C.2 Clone and Reproduce

To clone the repository and reproduce all analysis results locally:

```
git clone \
  https://github.com/daomuyang/Liesen-Gai-Causal-Inference-Final-Project.git
cd Liesen-Gai-Causal-Inference-Final-Project

python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install pandas numpy statsmodels scikit-learn matplotlib

MPLCONFIGDIR=.mplconfig python code/analysis.py
```

After successful execution, `output/summary.json` and PDFs in `figures/` match the text. Compile the report in `report/` with `pdflatex final_pj_en.tex` (twice). The analysis sample size is 68,392 after cleaning 70,000 raw records; do not mistake other UCI Heart Disease files in `data/` for the dataset used here.

References

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